Lundeen Cahilly

+1 (203) 780-1564 | lundeen.cahilly@gmail.com | github.com/lundeen06 | linkedin.com/in/lundeencahilly

Education

Stanford University — *Stanford, CA*
B.S. Computer Science; M.S. Aeronautics & Astronautics (Intended) — **Sep. 2024 – Jun. 2029**
- GPA: 3.93/4.00 | Courses: Mechanics & Special Relativity, Advanced Space Mechanics, Machine Learning, Decision Making Under Uncertainty, Special Operations Forces in an Era of Great Power Competition

Phillips Academy — *Andover, MA*
- GPA: 6.0/6.0 | Courses: Modern Physics, Astronomy Research, AP Physics C Mechanics & E&M, Art of Persuasion

Experience

K2 Space — *Torrance, CA*
GNC Intern — **Jun. 2026 – Sep. 2026**
- K2's First High-Orbit Momentum Dumping System: Owned end-to-end simulation and GNC development for K2's first in-house momentum dumping system for MEO/GEO/+. Will fly on the Trinity mission in Q1 2027 and all future missions. Wrote high-fidelity dynamics models, traded control architectures, assessed mechanism lifetime, implemented all flight control and estimation algorithms, and validated to flight standards
- New Trinity Guidance Mode: Developed algorithm to track Earth's horizon in elliptical orbits. Will fly on Trinity
- On-Orbit State Estimation Correction: Hardened attitude estimator against anomalous measurements seen on Gravitas
- Simulation Optimization: Identified and implemented opportunity to increase pointing accuracy specification by 15%

Stanford Student Space Initiative — *Stanford, CA*
Co-President — **Apr. 2026 – Present**
- Lead Stanford's largest engineering club (~300 members), overseeing satellite, rocketry, propulsion, biology, and high-altitude balloon programs alongside fundraising, recruiting, and external partnerships

Satellites Team Lead — **Apr. 2025 – Apr. 2026**
- Led 40+ engineers on the "SAMWISE" 2U CubeSat project, integrating autonomous GNC, an imaging payload, advanced over-the-air updates, and S-band communications
- Flight ADCS from Scratch: Over Summer 2025, spent >500hrs building SAMWISE's ADCS completely from scratch. Included writing 10+ drivers, quaternion/matrix libraries, an IGRF magnetic field model, control algorithms, desaturation, a multiplicative extended Kalman filter for state estimation, detumbling, and other pointing modes
- Launch Re-Manifest: Led crisis response negotiations with Exolaunch following an unprecedented bump from Transporter-15 launch date, securing the next-best launch window while strengthening relationship

Software & GNC Engineer — **Sep. 2024 – Mar. 2025**
- Wrote comprehensive 6-DOF attitude & orbit propagator with quaternion-based kinematics, perturbation torques, and world models to enable our team to develop GNC algorithms for the SAMWISE mission

Virtual Human Interaction Lab (VHIL) - PI Prof. Jeremy Bailenson — *Stanford, CA*
Research Assistant — **Jan. 2025 – Apr. 2025**
- Developed trackerless computer-vision and AI-based motion-tracking systems for multimodal XR studies

Dragon Group LLC — *Greenwich, CT*
Analyst — **Jun. 2017 – Aug. 2024**
- Tech-centric PE / GE / VC firm. Work with founders to accelerate global growth. Helped originate 8 portfolio positions to date, including xAI, Corsair Gaming, Elgato, Origin, SCUF, Epic Games, WNBA Portland Fire, and BTC
- xAI: Series B, Dec 2023. Member of core team leading Dragon's inaugural AI investment with truth maximization thesis
- Epic Games: Identified Unreal Engine's "hidden value" as a Real-Time 3D engine during 2017 Fortnite growth phase
- WNBA Portland Fire: Helped lead Dragon's second inaugural investment in women's professional sports

Fun Facts

Ran 50km ultra around Stanford's "Lake Lag". Been to 24 countries. Survived armed car chase. Ligado 501c3 co-founder